Spotify™

Distributed Systems

Jon Åslund
@jooon

TopSlideDeck

storage

consistent hashing

Spotify

Pitch

playlist

Spotify

Pitch

playlist

cassandra

Spotify

Pitch ck

playlist

structured key-value

store

Spotify

Pitch

playlist

high throughput

Spotify

playlist

choose redundancy and consistency level

Pitch

playlist

datacentre aware

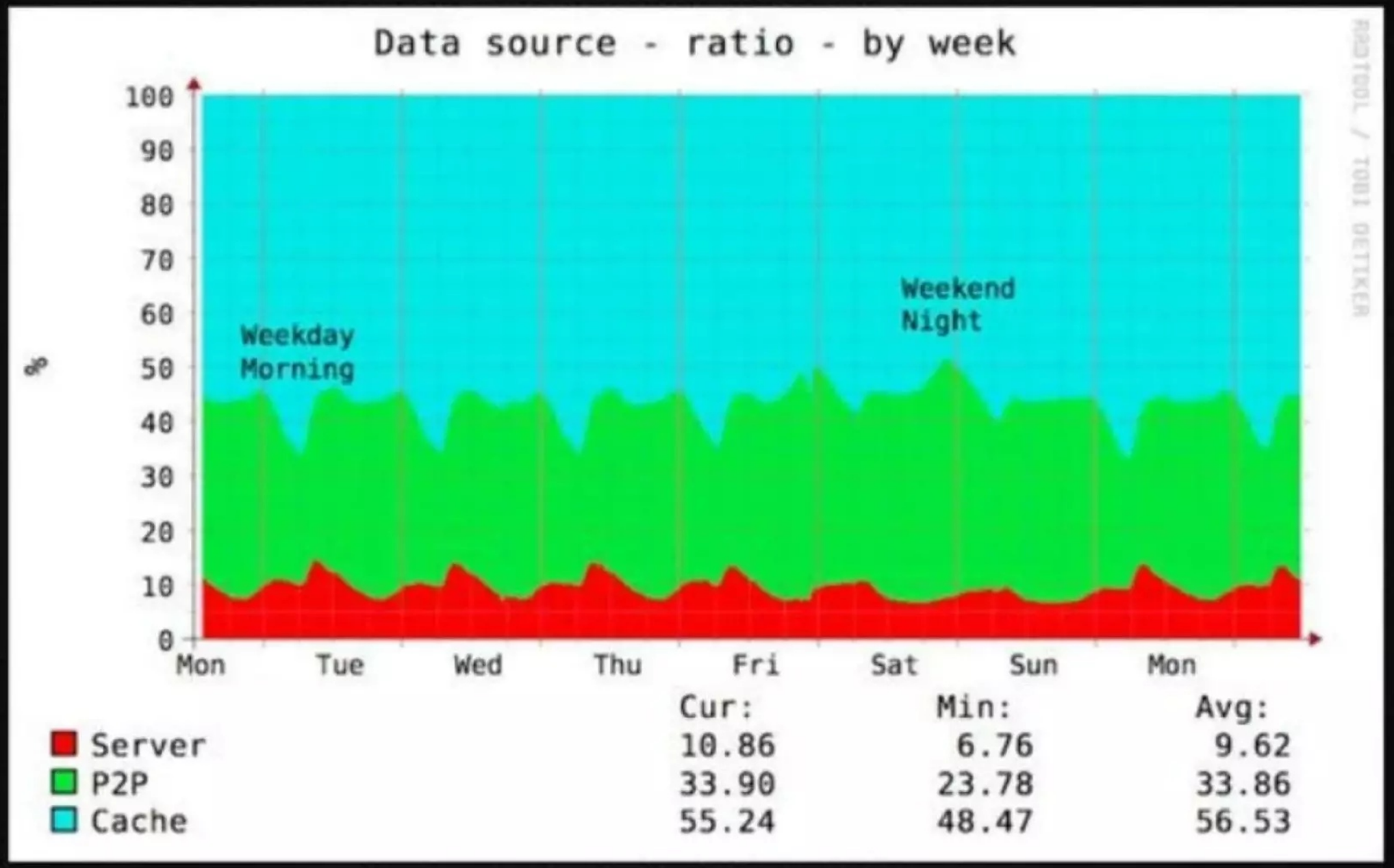
Spotify

Pitch

p2p

Spotify

Pitch


Data source - ratio - by week
RRDTOOL / TOBI OETIKER
100
90
80
70
60
50
40
30
20
10
0
%
Weekday
Morning
Weekend
Night
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Mon
Cur:
Min:
Avg:
Server
10.86
6.76
9.62
P2P
33.90
23.78
33.86
Cache
55.24
48.47
56.53

thank you

Spotify

Pitch

Master Thesis?

Spotify

Pitch

Yes, please!

Spotify

Pitch

or come and work for us

Spotify

Pitch

Oh, and we have cake as well! :D

spotify.com/jobs

Spotify

Who am I?

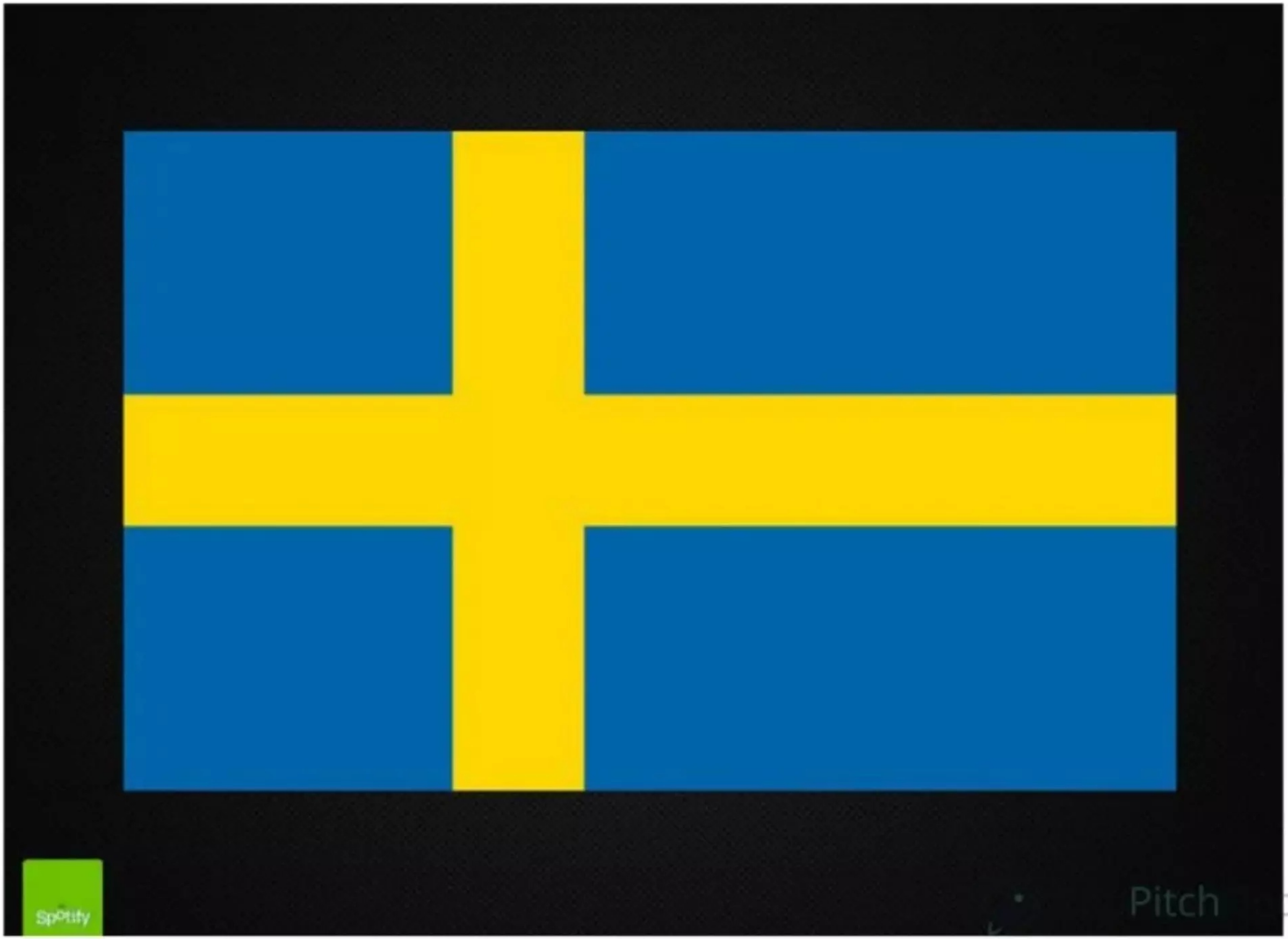
Spotify

Pitch

Jon Åslund

Spotify

Pitch


Spotify
Pitch

Computer Science

Spotify

Pitch

KTH

VETENSKAP
OCH KONST

Shakespeare Programming Language

Spotify

Pitch

Joined Spotify
Autumn 2006

Spotify

Pitch

April 2006

the idea

spotify

Pitch ck

Spotify™
Spotify

Spotify™
Spotify

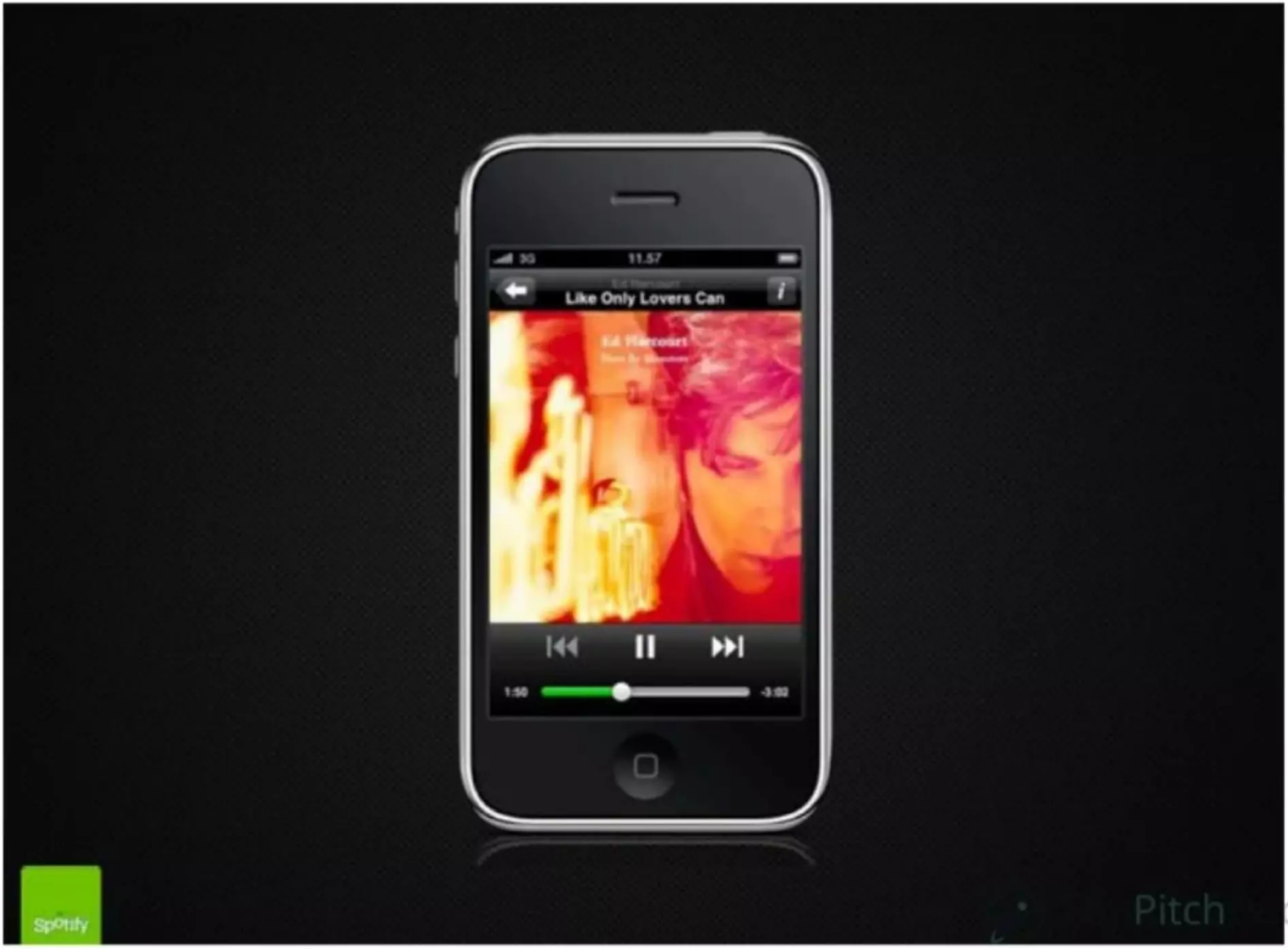
"music itself is going to become like running water or electricity"

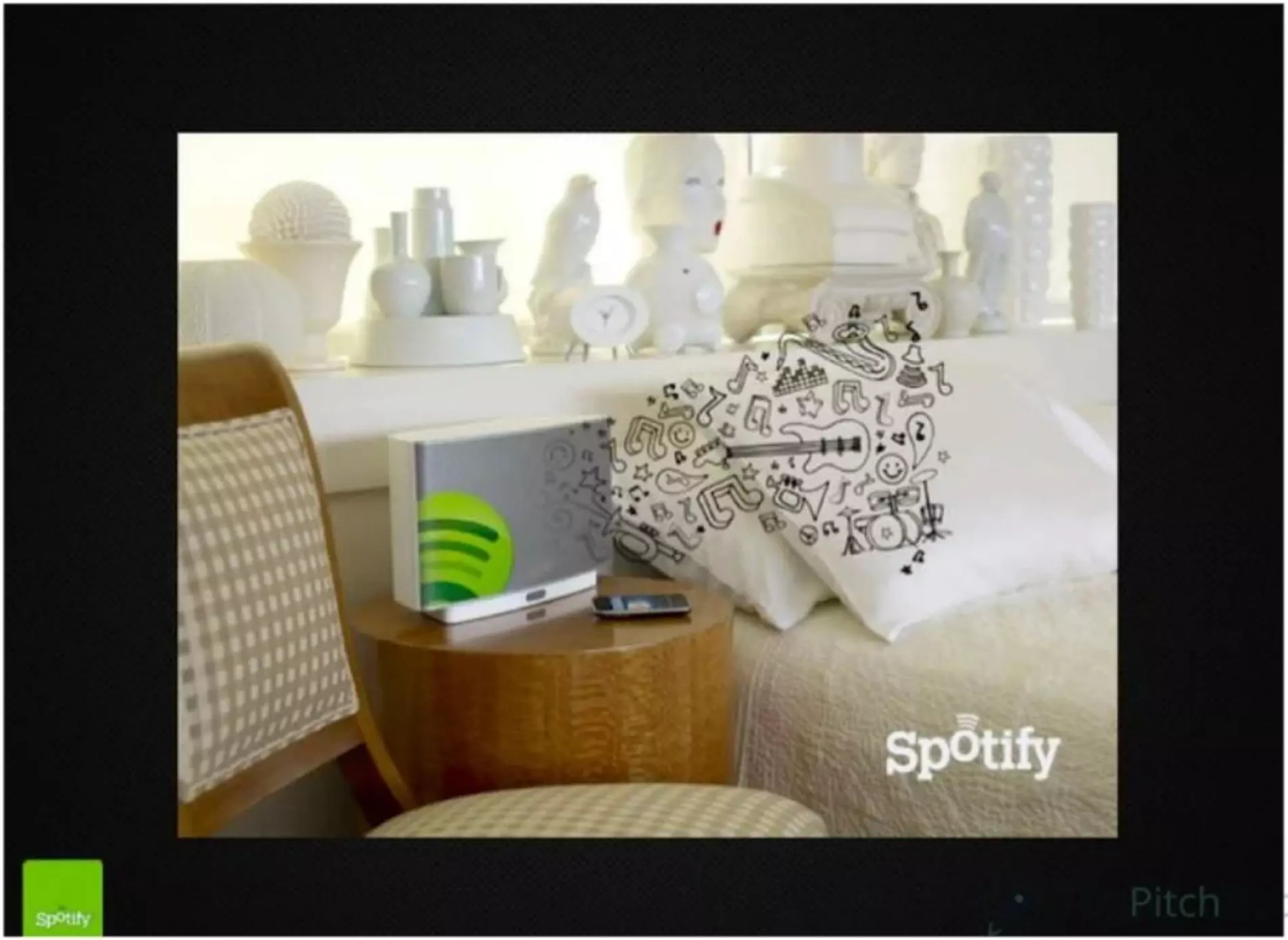
David Bowie, 2002

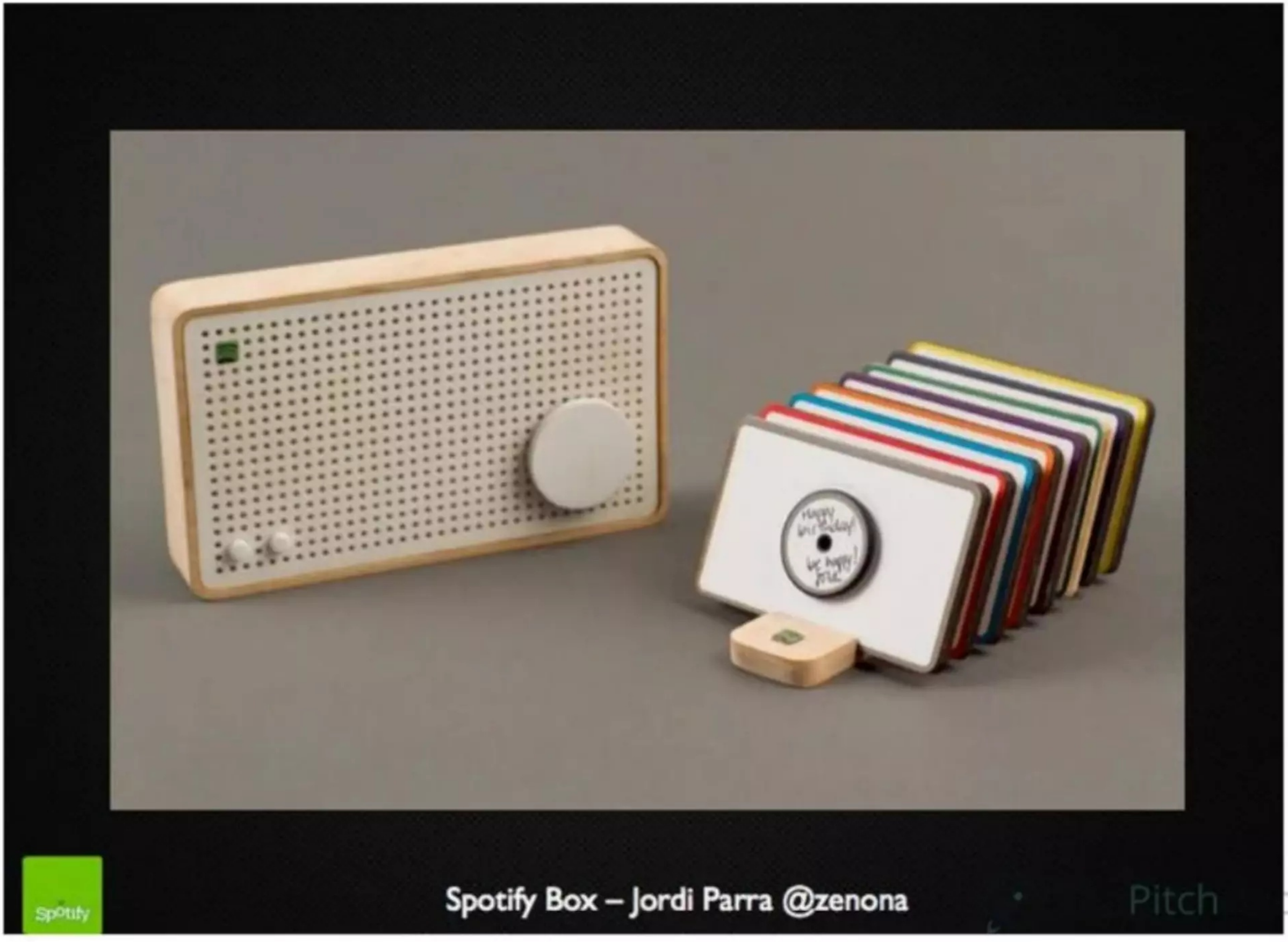
Spotify

Pitch

Spotify premium
What's new
Top lists
Feed
What's new?
More
People
Feed
Oskar Leijon: hedrar Guru med en liten nostalgilyssning
Jazzmatazz Volume 1
An album by Guru
Joachim Bengtsson: Obliterates ears and bugs alike
Basic Understanding
An album by Felix Krocher
Spotify: Also available today for Spotify Premium users in Norway and Sweden - the new Johnossi album, Mavericks, now available.
Spotify


3G
11.57
Like Only Lovers Can
1:50
-3:02
Spotify


Spotify



Spotify Box – Jordi Parra @zenona

February 2012

Spotify

Pitch

10+ million users

(active within previous 30 days)

Spotify

Pitch

15+ million tracks

Spotify

Pitch

500+ million playlists

Spotify

Pitch

1000+ servers

in 3 countries

Spotify

Pitch

Spotify is a distributed system

Spotify

Pitch

Fast

<200ms

Spotify

Pitch

Always working

Spotify

Pitch

How do you accomplish that?

Spotify

Pitch

Cheat

as much as possible

Spotify

Pitch

Fast

vs

Feels fast

Spotify

Pitch

Always working

vs

Pretend there is nothing wrong

Spotify

Pitch

Build to fail

Spotify

Pitch

Redundancy
(talk about waffles)

Spotify

Pitch

Error recovery

Spotify

Don't forget to test
your recovery code

Spotify

Pitch

Fire drill: Cut the atlantic cable

Spotify

Pitch

Netflix
Chaos Monkey

Spotify

Pitch

"On January 15, 1990, AT&T's long-distance telephone switching system crashed."

Spotify

Pitch

Isolation

Spotify

Pitch

Getting the right tool for the job

Spotify

SPELAR NU: Bosse - fin låt 2.13

SÖK

MITT BIBLIOTEK

SÖK

MODER-BIBLIO-TEKET

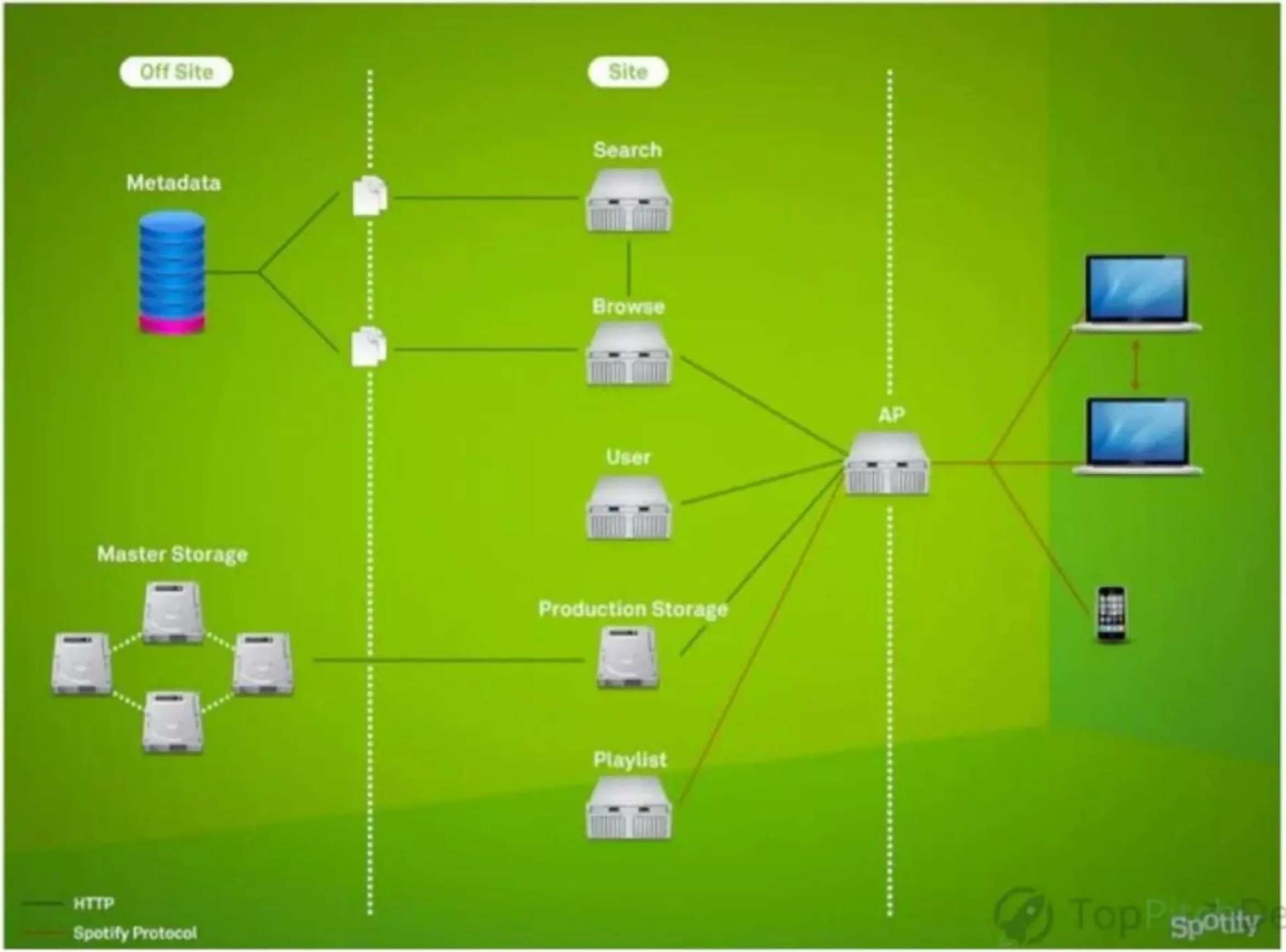
Listar & filer

Spellista

LOGO

REKLAM/Spotlar

MER REKLAM


Off Site
Site
Search
Metadata
Browse
AP
User
Master Storage
Production Storage
Playlist
HTTP
Spotify Protocol
Spotify

Consistency

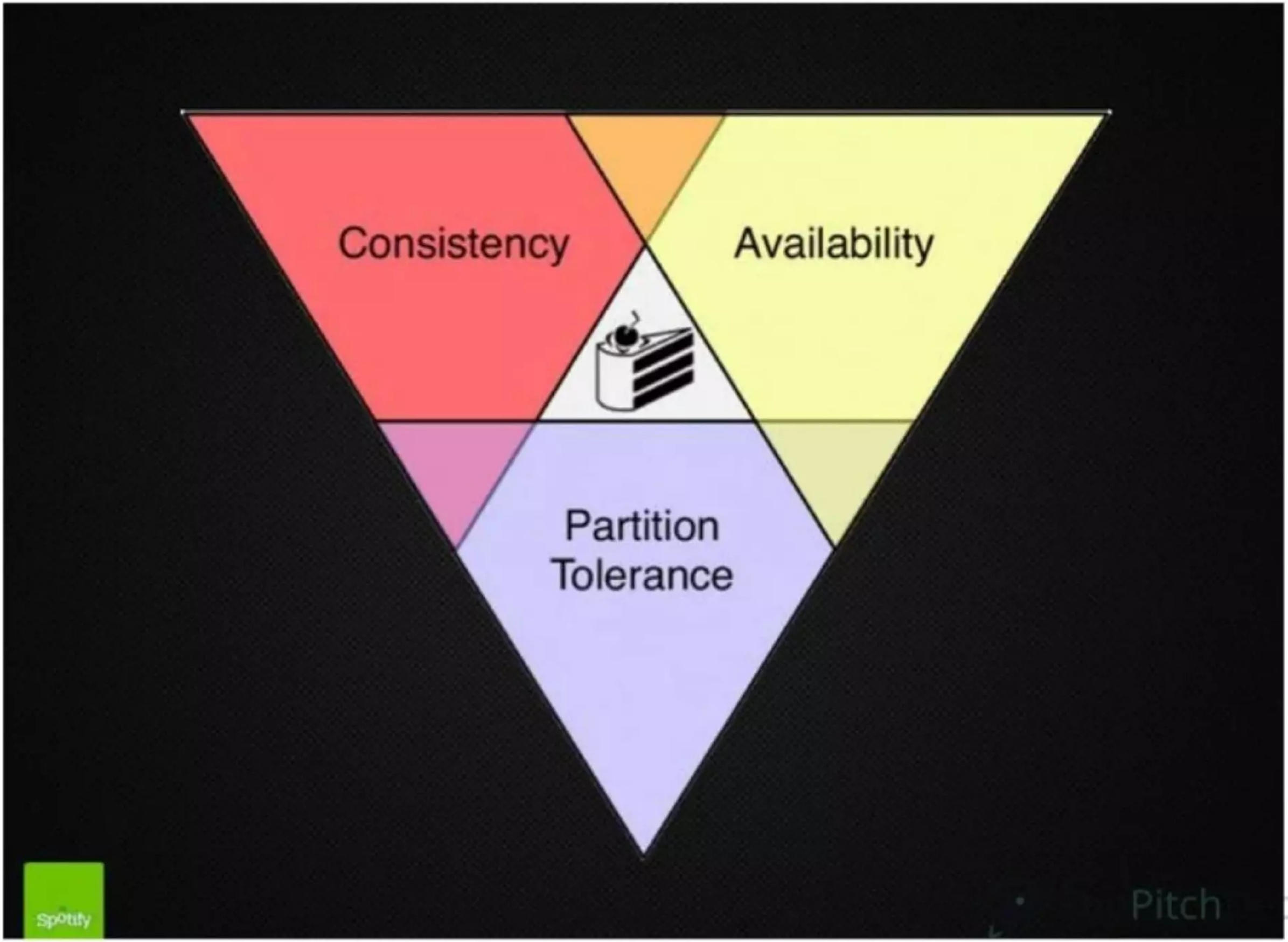
Availability

Partition tolerance

http://ksat.me/a-plain-english-introduction-to-cap-theorem/

Spotify

Pitch


Consistency
Availability
Partition
Tolerance
Spotify

user

Spotify

Pitch

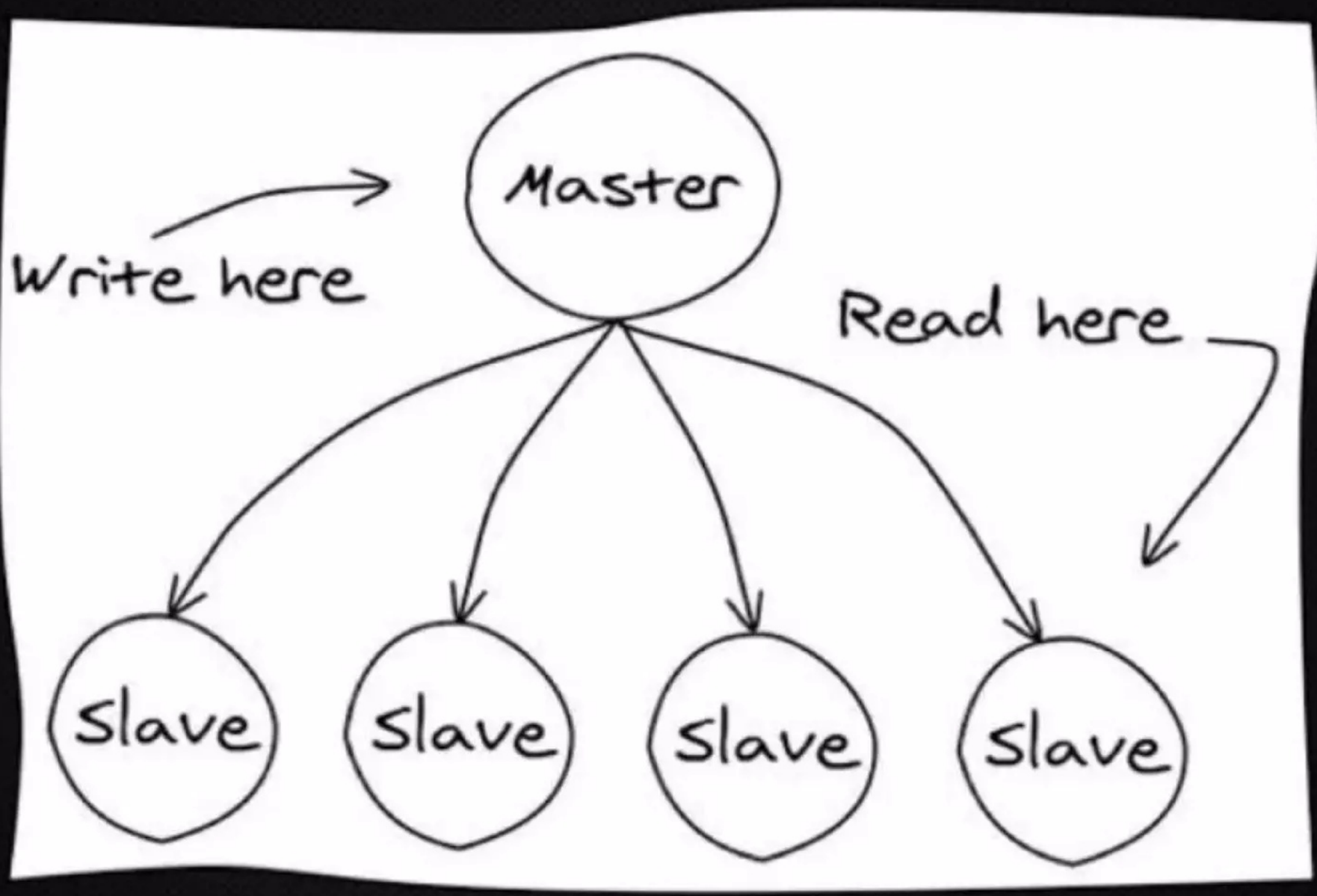
user

PostgreSQL

Sp0tify

Pitch

user



Spotify

search

Spotify

Pitch

search

Lucene index in memory

Spotify

Pitch

search

replaced daily

Spotify

Pitch

search

read only

Spotify

Pitch

search

scales horizontally

Spotify

Pitch

search

sort of

Spotify

Pitch

search

How do you replace an index in memory when you are using all of it?

Spotify

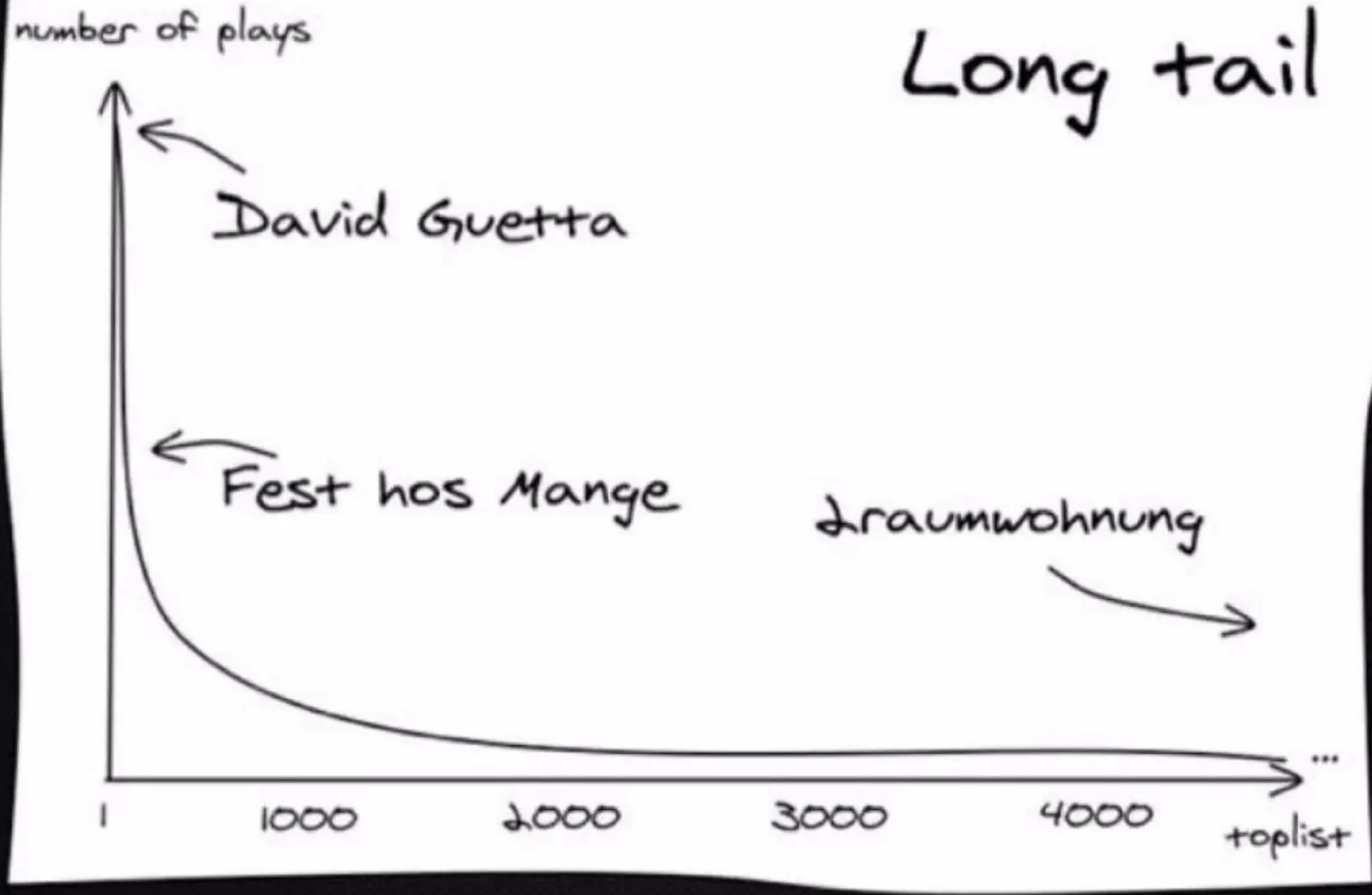
Pitch ck

storage

Spotify

Pitch

storage



Spotify

Pitch

storage

lru cache in nginx

Spotify

storage

better use of memory with sharding

Spotify

Pitch

Storage

0-24

75-99

25-49

50-74

TopDeck